Development of Temperature Resistant high-strength Al alloys

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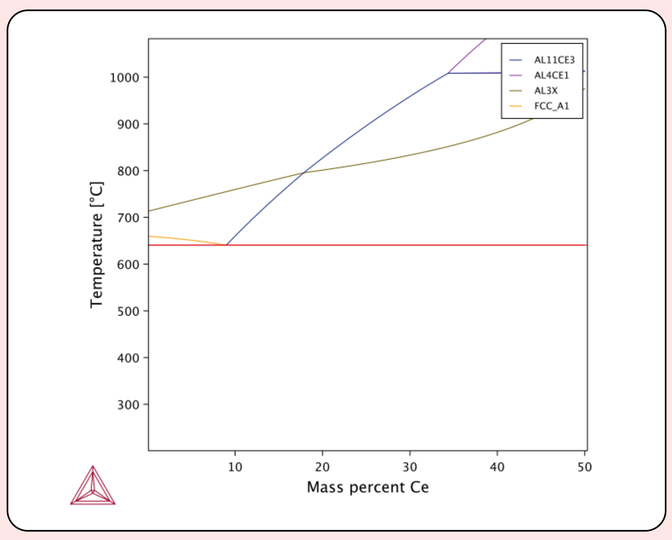
Context

Conventional aluminium alloys soften rapidly above 150–250 °C, limiting their usefulness in high-temperature environments. This work explores alloying strategies to enhance thermal stability while retaining high strength.

Idea

We investigate an aluminium alloy using ~10% cerium to create a thermally stable eutectic, and 0.2–1% scandium to refine the microstructure and enhance high-temperature strength **(300-350 °C).**



We study two systems :

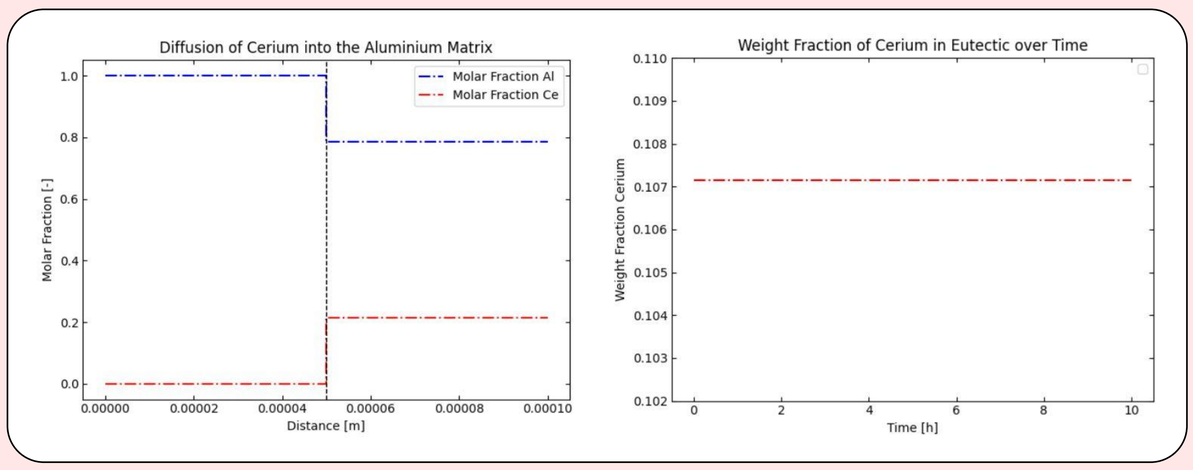
- The eutectic : **$Al_{11}Ce_3$**
- The precipitate : **Al_3Sc**

Aluminium Cerium Alloys

Diffusion behavior of **$Al_{11}Ce_3$** (eutectic phase) :

• 350 °C for 2 days • 300 °C for 2 days

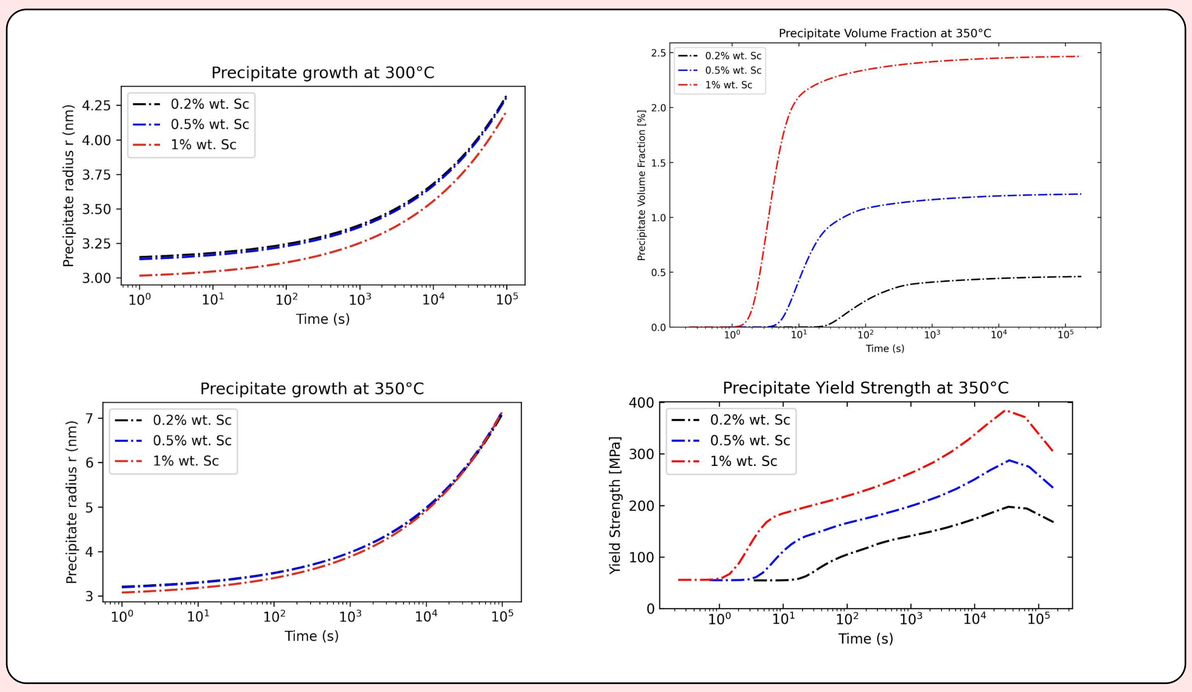
⇢ Effect of temperature and treatment duration



Results : $Al_{11}Ce_3$ shows almost no diffusion at 350 °C, , or 300 °C, demonstrating exceptional thermal stability and confirming its suitability as a high-temperature strengthening eutectic.

Aluminium Scandium Alloys

1/ Precipitation behavior of Al_3Sc during heat treatment and under high-temperature service conditions. Attention was paid to the evolution of precipitate size and stability, as well as to their contribution to precipitation strengthening.



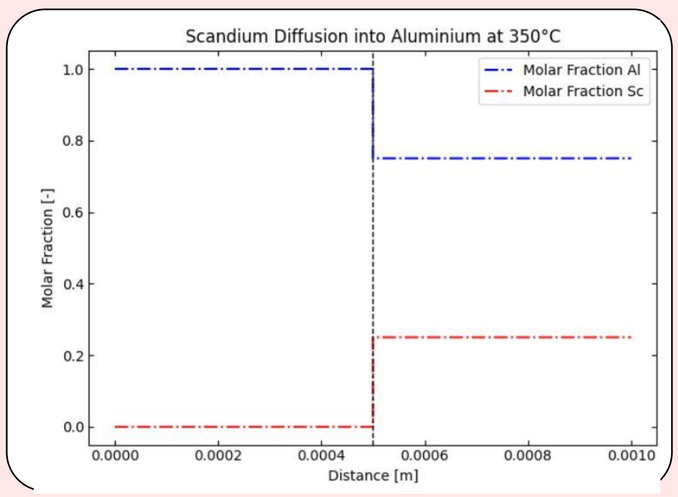
Results :

- Heat treatment (350 °C/2day) : performant strengthening
- Long-term exposure (300 °C/2day) : superior stability. Higher Sc contents improve strengthening but high material cost.

2/ Diffusion behavior of **Al_3Sc** :

• 350 °C for 2 days • 300 °C for 2 days

⇢ Effect of temperature and treatment duration



Results :
Molar fraction profiles show stable diffusion, low Al–Sc interdiffusion and confirming the high thermal stability of the Al_3Sc phase at the precipitate–matrix interface.

Conclusion

- **Ce (~10%) forms the $Al_{11}Ce_3$ eutectic**, providing excellent thermal stability.
- **Sc (0.2–1%) forms Al_3Sc precipitates**, refining and strengthening the matrix.
- **The combined Ce–Sc mechanisms enhance aluminium's high-temperature performance.**
- **Al_3Sc remains stable at 300 °C but becomes limiting at 350 °C** due to precipitate coarsening.
- **Al–Ce–Sc alloys are thus promising for moderate high-temperature applications**

Composition	Model time (s)	Yield strength (MPa)		Decrease from peak (%)	
		300 °C	350 °C	300 °C	350 °C
0.2 wt% Sc	10^3	166.9	159.7	0.7	5.0
	10^4	163.9	144.5	2.5	12.8
	10^5	153.9	128.1	8.4	23.8
0.5 wt% Sc	10^3	278.3	266.3	0.0	4.3
	10^4	266.3	237.9	4.3	14.5
	10^5	237.9	177.6	14.5	36.2
1.0 wt% Sc	10^3	378.5	365.8	0.0	3.3
	10^4	365.8	323.0	3.3	14.7
	10^5	365.8	228.3	3.3	39.7

EPFL

Thermo-Calc Software